

# CDISC 360i: Automating from Study Design to Submission

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# Agenda

- Automation with 360i
- Achievements and Insights
- The Future



## Speaker



**Darren Moreland**  
Director, Digital  
Transformation  
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- Darren Moreland is the Director of Digital Transformation at CDISC with over 20 years of technical experience across multiple industries. He has spent the last 10 years working with pharmaceutical companies simplifying clinical trial design, data management and SDTM submissions.

# The Problem

- Time-consuming manual processes
- Data duplication
- Operational silos
- High effort to create submission datasets
- Gaps in data traceability







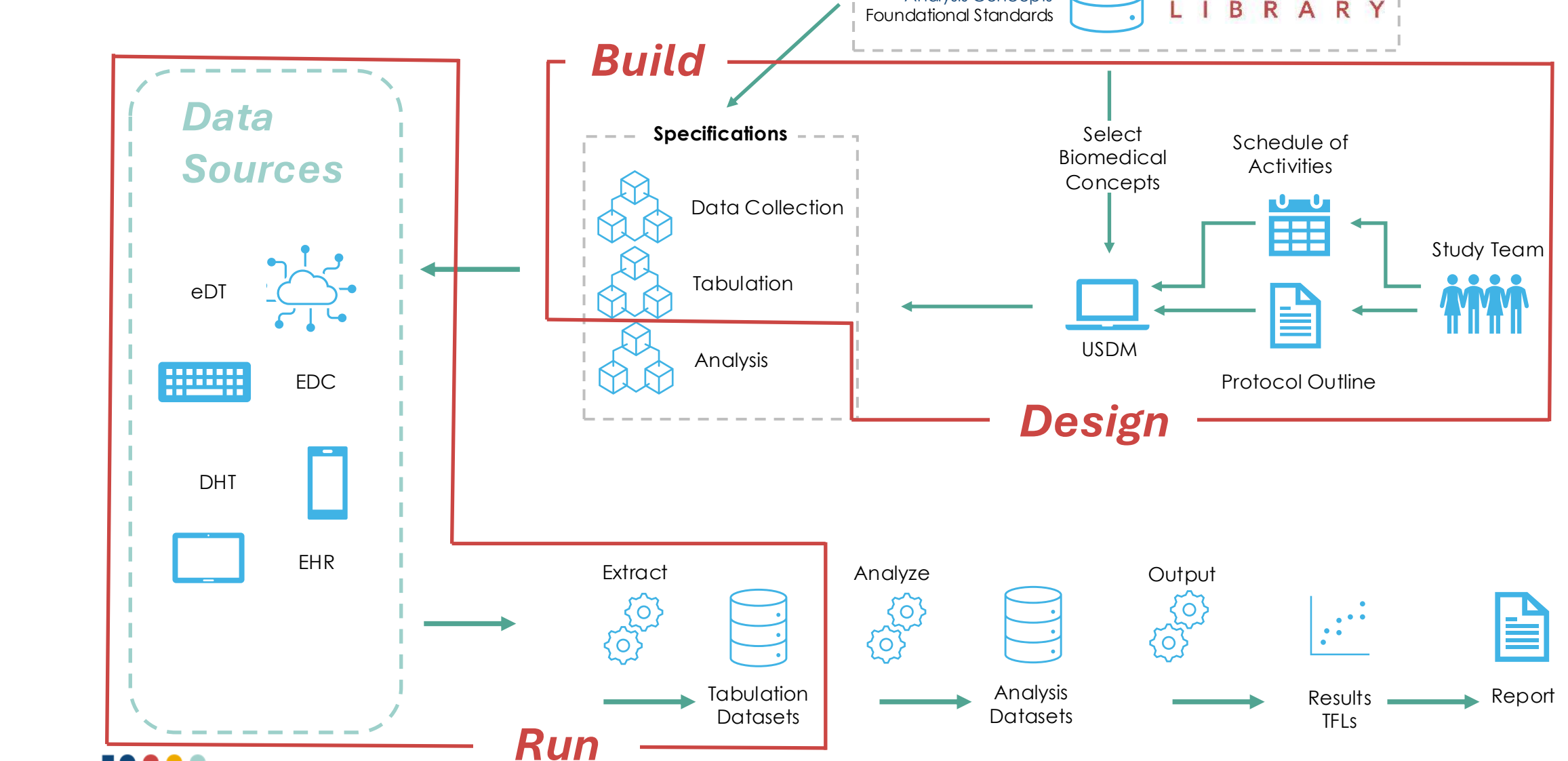
# The Vision: CDISC 360i

## Strategic Goal

A multi-year strategy transitioning the industry from static, document-based standards and processes to a dynamic, digital data flow beginning with digital protocols



# 360i Phase 1 Focus



# USDM

- Single authoritative source
- Machine-readable, structured study design
- Reliable exchange of data across systems
- Protocol concepts linked to downstream models using semantic layer
  - Shared, interoperable metadata layer linking CDISC standards
  - Biomedical Concepts linked to the Schedule of Activities



# Schedule of Activities

Matrix: Main Timeline

|                                                 | Epoch:->                                          | Screening   |             | Treatment 1 |            |            | Treatment 2 |            |            |            |             |            |             |            |             |            | Treatment 3 |
|-------------------------------------------------|---------------------------------------------------|-------------|-------------|-------------|------------|------------|-------------|------------|------------|------------|-------------|------------|-------------|------------|-------------|------------|-------------|
|                                                 | Encounter:->                                      | Screening 1 | Screening 2 | Baseline    | Week 2     | Week 4     | Week 6      | Week 8     | Week 8     | Week 8     | Week 12     | Week 12    | Week 16     | Week 16    | Week 20     | Week 20    | Week 24     |
|                                                 | Instance:->                                       | Screen 1    | Screen 2    | Dose        | Week 2     | Week 4     | Week 6      | Week 8     | Week 8 NPI | Week 12    | Week 12 NPI | Week 16    | Week 16 NPI | Week 20    | Week 20 NPI | Week 24    |             |
|                                                 | Study Day:->                                      |             |             | 1.0         | 14.0       | 28.0       | 42.0        | 56.0       | 56.0       | 84.0       | 84.0        | 112.0      | 112.0       | 140.0      | 140.0       | 168.0      |             |
|                                                 | Timing:->                                         | Screening   | Pre-dose    | Dosing      | Week 2     | Week 4     | Week 6      | Week 8     | Week 8     | Week 12    | Week 12     | Week 16    | Week 16     | Week 20    | Week 20     | Week 24    |             |
|                                                 | Visit Window:->                                   |             | -4.0 hours  |             | -3..3 days | -3..3 days | -3..3 days  | -3..3 days | -3..3 days | -4..4 days | -4..4 days  | -4..4 days | -4..4 days  | -4..4 days | -4..4 days  | -4..4 days |             |
| Activity                                        | Concepts                                          |             |             |             |            |            |             |            |            |            |             |            |             |            |             |            |             |
| Informed Consent                                | Concepts 1 ▶                                      | X           |             |             |            |            |             |            |            |            |             |            |             |            |             |            |             |
| Inclusion/Exclusion Criteria                    | Concepts 3 ▶                                      | X           |             |             |            |            |             |            |            |            |             |            |             |            |             |            |             |
| Patient Number assigned                         | Concepts 1 ▶                                      | X           |             |             |            |            |             |            |            |            |             |            |             |            |             |            |             |
| Demographics                                    | Concepts 5 ▶                                      | X           |             |             |            |            |             |            |            |            |             |            |             |            |             |            |             |
| Education                                       | Concepts 2 ▶                                      | X           |             |             |            |            |             |            |            |            |             |            |             |            |             |            |             |
| Hachinski                                       | Concepts 10 ▼                                     |             |             |             |            |            |             |            |            |            |             |            |             |            |             |            |             |
|                                                 | [S] Abrupt Onset (C112618)                        |             |             |             |            |            |             |            |            |            |             |            |             |            |             |            |             |
|                                                 | [S] Stepwise Deterioration (C112619)              |             |             |             |            |            |             |            |            |            |             |            |             |            |             |            |             |
|                                                 | [S] Emotional Incontinence (C112620)              |             |             |             |            |            |             |            |            |            |             |            |             |            |             |            |             |
|                                                 | [S] History or Presence of Hypertension (C112621) |             |             |             |            |            |             |            |            |            |             |            |             |            |             |            |             |
|                                                 | [S] History of Stroke (C112622)                   | X           |             |             |            |            |             |            |            |            |             |            |             |            |             |            |             |
|                                                 | [S] Focal Neurological Symptoms (C112623)         |             |             |             |            |            |             |            |            |            |             |            |             |            |             |            |             |
|                                                 | [S] Focal Neurological Signs (C112624)            |             |             |             |            |            |             |            |            |            |             |            |             |            |             |            |             |
|                                                 | [S] Somatic Complaints (C112813)                  |             |             |             |            |            |             |            |            |            |             |            |             |            |             |            |             |
| [S] Total Score (C112625)                       |                                                   |             |             |             |            |            |             |            |            |            |             |            |             |            |             |            |             |
| [S] Modified Hachinski Ischemic Scale (C112814) |                                                   |             |             |             |            |            |             |            |            |            |             |            |             |            |             |            |             |
| MMSE                                            | Concepts ▶                                        | X           |             |             |            |            |             |            |            |            |             |            |             |            |             |            |             |
| Physical examination                            | Concepts ▶                                        | X           |             |             |            |            |             |            |            |            |             |            |             |            |             |            |             |
| Medical History                                 | Concepts 2 ▶                                      | X           |             |             |            |            |             |            |            |            |             |            |             |            |             |            |             |
| Habits                                          | Concepts 3 ▶                                      | X           |             |             |            |            |             |            |            |            |             |            |             |            |             |            |             |
| Chest X-ray                                     | Concepts 1 ▶                                      | X           |             |             |            |            |             |            |            |            |             |            |             |            |             |            |             |
| Apo E genotyping                                | Concepts ▶                                        |             |             |             | X          |            |             |            |            |            |             |            |             |            |             |            |             |
| Patient randomized                              | Concepts 1 ▶                                      |             |             | X           |            |            |             |            |            |            |             |            |             |            |             |            |             |
| Vital Signs                                     | Concepts 7 ▼                                      |             |             |             |            |            |             |            |            |            |             |            |             |            |             |            |             |
|                                                 | Body Temperature                                  |             |             |             |            |            |             |            |            |            |             |            |             |            |             |            |             |
|                                                 | Body Weight                                       |             |             |             |            |            |             |            |            |            |             |            |             |            |             |            |             |
|                                                 | Body Height                                       |             |             |             |            |            |             |            |            |            |             |            |             |            |             |            |             |
|                                                 | Pulse Rate                                        | X           | X           | X           | X          | X          | X           | X          | X          | X          |             | X          |             | X          |             | X          |             |
|                                                 | Respiratory Rate                                  |             |             |             |            |            |             |            |            |            |             |            |             |            |             |            |             |
|                                                 | Body Mass Index                                   |             |             |             |            |            |             |            |            |            |             |            |             |            |             |            |             |
|                                                 | Blood Pressure                                    |             |             |             |            |            |             |            |            |            |             |            |             |            |             |            |             |
| Aspirator FOC                                   | Concepts ▶                                        |             |             |             |            |            |             |            |            |            |             |            |             |            |             |            |             |



## 360i Architecture

- USDM as single source of truth
- Enables early asset generation
  - Trial Design Domains
  - define.xml
  - CRFs/aCRFs
  - SDTM shell datasets
- Binds protocol concepts defined in USDM to SDTM datasets
- Data persistence and traceability



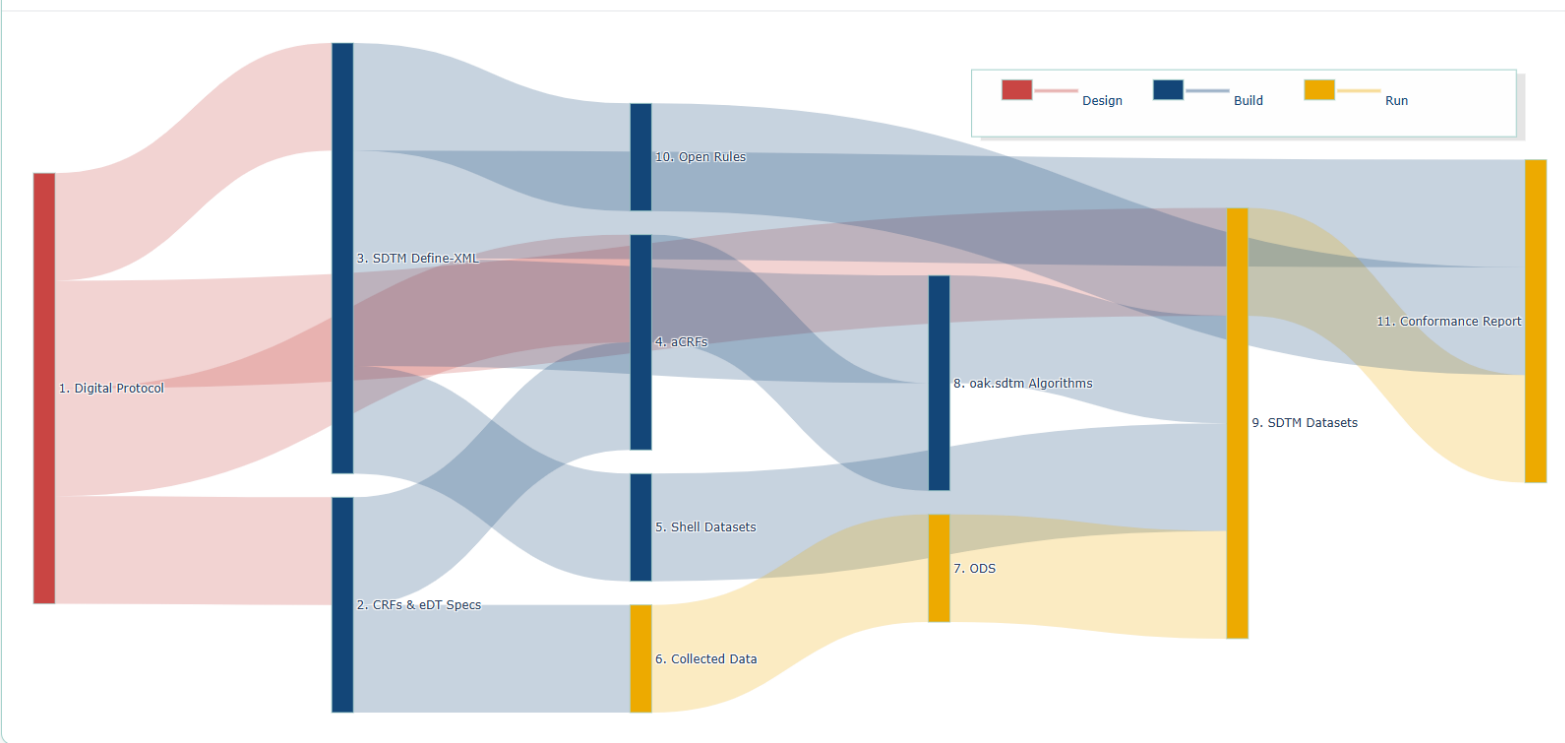
# Phase 1

- Integration of independent open-source projects into automated pipeline from UDSM to SDTM
  - CRFs & aCRFs in ODM-XML, Dataset-JSON
  - Define.xml
  - SDTM shell datasets
  - SDTM Trial Design Domains
  - SDTM datasets
  - CORE Conformance reports



# Design and Build

360i Flow from Digital Protocol





## Methods

- Data store for asset storage and versioning
- Author and validate USDM study design
- Python & R orchestration to create SDTM
- CDISC Open Rules Engine (CORE) for conformance validation
- Google Colab notebooks for accessibility
  - Explanation and instructions inline with code





## Early Asset Generation

- Complete USDM allows earlier creation of study assets
  - Trial Design Domains
  - define.xml
  - SDTM Shell datasets
  - CRFs & aCRFs in multiple formats
- Can address issues earlier in the study lifecycle







## Tools & Technology

- Use open-source libraries and existing solutions
  - Python & R
  - SQLite
  - CORE validation
  - CDISC Library
    - Data Set Specializations
    - Biomedical Concepts
    - SDTM Controlled Terminology
  - [GitHub](#)

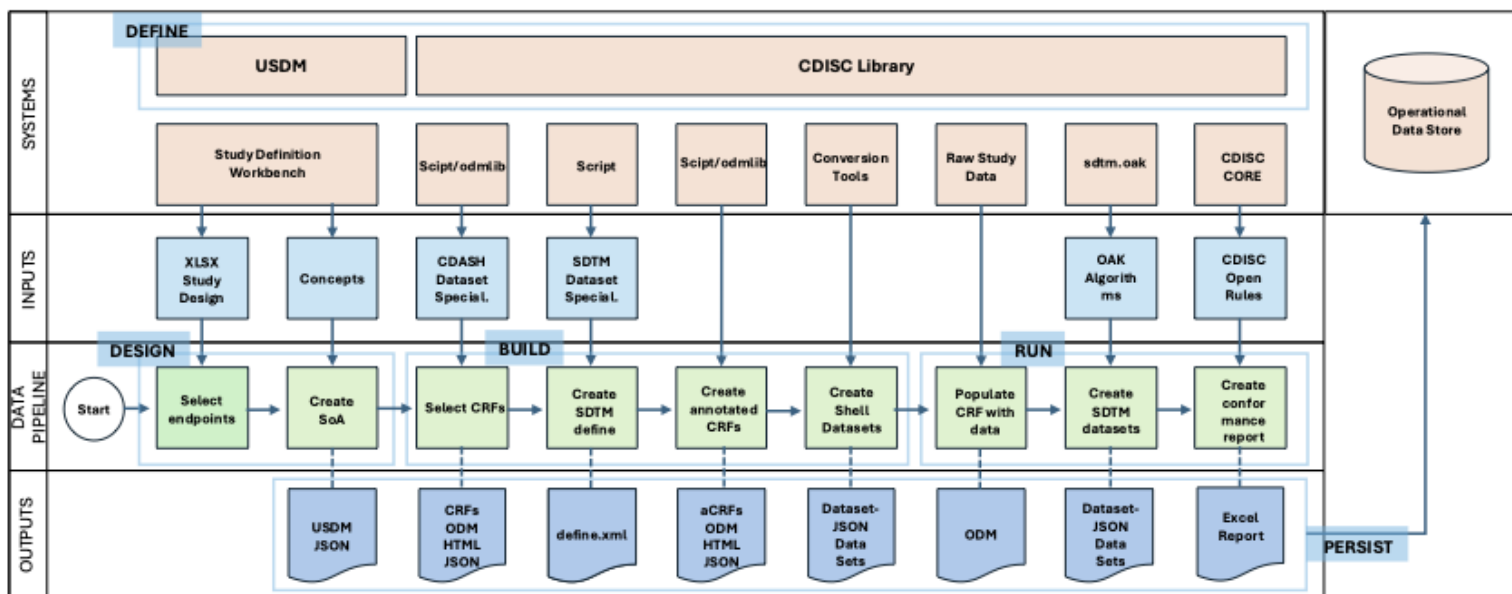


## Results

- Automated pipeline integrating independent projects creating SDTM from USDM
  - CRFs & aCRFs – ODM, Dataset-JSON, HTML
  - Trial Design Domains – CSV, XPT, Dataset-JSON
  - define.xml
  - SDTM domains – CSV, XPT, Dataset-JSON
  - All raw data and output stored to operational data store (ODS)



# 360i Phase 1: Generating SDTM



## Lessons Learned

- USDM can drive downstream automation using interoperable metadata
- CORE checks confirm conformance early in the development lifecycle
- Open-source
  - Ensure community involvement
  - Independent workstreams
  - CDISC standards are tool agnostic; allows contributors to focus on solutions over standards using whatever tools meet their needs





# The Future

- Phase 2 of the CDISC 360i initiative
  - ADaM and Analysis Result Sets (ARS)
  - Expand concepts for analysis and derivation concepts
  - Define-JSON
  - SoA Workbench
  - Automate impact analysis for protocol amendments





## Conclusion

- Structured, interoperable metadata in USDM digital protocol can be used to automate the creation of SDTM
- Focus on solutions over standards to ensure independent development and wider engagement with the open-source community





## GitHub References

| Purpose                       | Location                                                                                                                                                                                          |
|-------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| CDISC CORE Rules Engine       | <a href="https://github.com/cdisc-org/cdisc-rules-engine">https://github.com/cdisc-org/cdisc-rules-engine</a>                                                                                     |
| CRF creation                  | <a href="https://github.com/lexjansen/cdisc360i-pocs">https://github.com/lexjansen/cdisc360i-pocs</a>                                                                                             |
| Define-XML creation           | <a href="https://github.com/swhume/template2define">https://github.com/swhume/template2define</a>                                                                                                 |
| Define-XML template creation  | <a href="https://github.com/dostiep/360i">https://github.com/dostiep/360i</a>                                                                                                                     |
| Google Colab Notebooks        | <a href="https://github.com/cdisc-org/cdisc-360i-notebooks">https://github.com/cdisc-org/cdisc-360i-notebooks</a>                                                                                 |
| Open Study Builder            | <a href="https://gitlab.com/Novo-Nordisk/nn-public/openstudybuilder/OpenStudyBuilder-Solution">https://gitlab.com/Novo-Nordisk/nn-public/openstudybuilder/OpenStudyBuilder-Solution</a>           |
| Raw subject data              | <a href="https://github.com/alidootson/UpdatedCDISCPilotData/tree/main/UpdatedCDISCPilotData/CDASH">https://github.com/alidootson/UpdatedCDISCPilotData/tree/main/UpdatedCDISCPilotData/CDASH</a> |
| Study Definition Workbench    | <a href="https://github.com/data4knowledge/study_definitions_workbench">https://github.com/data4knowledge/study_definitions_workbench</a>                                                         |
| Study USDM documents          | <a href="https://github.com/cdisc-org/360i/tree/main/data/protocol/LZZT/usdm">https://github.com/cdisc-org/360i/tree/main/data/protocol/LZZT/usdm</a>                                             |
| Trial Design Dataset creation | <a href="https://github.com/pendingintent/cdisc-usdm-utils">https://github.com/pendingintent/cdisc-usdm-utils</a>                                                                                 |
| USDM validation utility       | <a href="https://github.com/pendingintent/cdisc-json-validation">https://github.com/pendingintent/cdisc-json-validation</a>                                                                       |

*A list of the projects and corresponding GitHub repositories integrated to create the automation*

# Using GitHub

**The GitHub repositories used in the demo are public and use open-source libraries**

The process for using these libraries is consistent:

- Fork the original public repository into personal GitHub repository
- Create new branch in personal GitHub repository
- Edit code to meet individual requirements, add features or fix issues
- Commit changes to personal branch
- If proposing change to parent (original) branch to share with wider community:
  - Raise a Pull Request with original repository owner
  - Original repository owner will review the Pull Request and work with requestor to evaluate and test Pull Request
  - Merge new code in Pull Request with original parent branch





## Accessing the 360i Notebook

The notebook displayed during this demonstration can be found here:

<https://github.com/cdisc-org/cdisc-360i-notebooks>

- Development of the notebook complete
- Development in other repos may result in some utilities not working
- Raise any issues encountered while using notebooks in GitHub







## We need you

- Be part of the team to drive this forward
  - We need committed contributors AND...companies that support the goals
  - Click on QR Code below to volunteer
  - Make sure to select 360i from the list!
- 
- <https://www.cdisc.org/volunteer>
  - ✓ Complete CDISC Volunteer Form
  - ✓ Complete 360i Contributor Survey



**Thank You!**



# Questions?



